Medvale Internal Medicine

Chapter 2D: Pleural Disease and Pneumothorax

Original pulmonary medicine study chapter for clinical reasoning, inpatient management, and graph-RAG concept extraction.

Chapter map
This segment covers pleural effusion, Light criteria, thoracentesis, parapneumonic effusion, empyema, pneumothorax, and tension pneumothorax. It builds directly on Chapter 2C because pneumonia can extend into the pleural space, and it prepares for later pulmonary oncology because malignancy is a common cause of recurrent exudative effusion.

Why this matters clinically

Pleural disease is one of the most common reasons that a patient with pulmonary symptoms suddenly becomes an inpatient problem. A small effusion may be an incidental clue to congestive heart failure, cirrhosis, nephrotic syndrome, pneumonia, malignancy, pulmonary embolism, or autoimmune disease. A large effusion can mechanically compress lung tissue, reduce tidal volume, worsen V/Q mismatch, and convert mild dyspnea into hypoxemic respiratory failure. A pneumothorax can be benign enough for observation or dangerous enough to cause obstructive shock before an imaging study is obtained. The core skill is therefore not memorizing a list of pleural diagnoses; it is linking physiology to the bedside decision: observe, tap, drain, decompress, or operate.

For internal medicine exams and real clinical care, pleural questions often present as a patient with dyspnea, pleuritic chest pain, decreased breath sounds, dullness or hyperresonance to percussion, a shifted trachea, or a new opacity on chest imaging. The key decision points are numeric: Light criteria thresholds, pleural fluid pH near 7.20, glucose near 40 to 60 mg/dL, pleural fluid thickness around 10 mm on decubitus imaging, recurrence risk after spontaneous pneumothorax, and the timing of urgent decompression. These values should not be memorized in isolation. Each number represents a physiologic transition point: from pressure imbalance to inflammation, from uncomplicated infection to complicated infection, from tolerable pleural air to dangerous hemodynamic compromise.

Core illness scripts

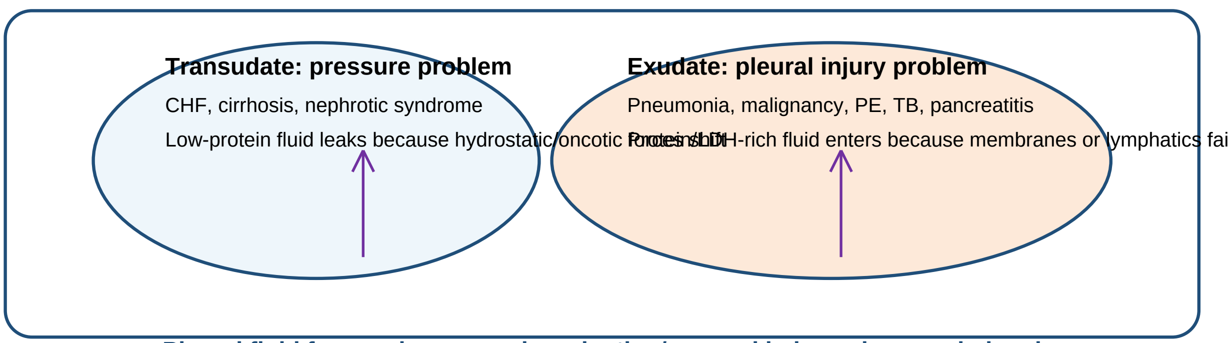
Syndrome	Prototype presentation	Mechanism	First management question
Transudative pleural effusion	Bilateral or right-predominant effusions in CHF/cirrhosis/nephrotic syndrome; dyspnea may improve with diuresis	Normal pleura with altered hydrostatic or oncotic pressure	Is the story classic enough to treat the cause first, or is thoracentesis needed because the effusion is unilateral, febrile, painful, large, or atypical?
Exudative pleural effusion	Fever, pleuritic pain, unilateral effusion, malignancy clues, PE symptoms, or inflammatory disease	Injured pleura, increased permeability, impaired lymphatic drainage, or local infection	What is the cause, and does the fluid require drainage rather than diagnostic sampling alone?
Complicated parapneumonic effusion/empyema	Pneumonia plus persistent fever, pleuritic pain, loculated fluid, low pH, low glucose, high LDH, or pus	Bacterial invasion and neutrophilic inflammation in pleural space	Antibiotics alone or source control with chest tube/VATS?
Primary spontaneous pneumothorax	Sudden unilateral pleuritic pain and dyspnea in tall, thin, young patient without known lung disease	Rupture of subpleural bleb allows air into pleural space	Is the patient stable and minimally symptomatic, or does air need removal?

Syndrome	Prototype presentation	Mechanism	First management question
Secondary spontaneous pneumothorax	COPD, cystic fibrosis, ILD, lung cancer, or pneumonia patient with acute dyspnea	Pleural air occurs in patient with limited pulmonary reserve	Most require intervention because physiologic reserve is low.
Tension pneumothorax	Hypotension, severe dyspnea, JVD, unilateral absent breath sounds, tracheal deviation late	One-way valve air trapping raises intrathoracic pressure, collapses lung, reduces venous return	Do not wait for CXR; decompress immediately.

Pathophysiology: what the pleural space normally does

The pleural space is a potential space between visceral pleura on the lung and parietal pleura on the chest wall. Under normal conditions it contains only a thin lubricating fluid layer. The negative intrapleural pressure helps keep the lung expanded against the chest wall. Disease enters this system through three broad mechanisms. First, fluid can accumulate because hydrostatic pressure rises or oncotic pressure falls, producing a transudate. Second, fluid can accumulate because the pleura itself is inflamed, infected, infiltrated by tumor, or poorly drained by lymphatics, producing an exudate. Third, air can enter the space, uncoupling the lung from the chest wall and producing pneumothorax.

A simple way to link pleural disease to the physical exam is to ask whether the pleural space is filled with fluid or air. Fluid blocks sound transmission and makes percussion dull; air blocks sound transmission and makes percussion hyperresonant. Both can reduce breath sounds. Both can cause dyspnea by reducing effective ventilated lung volume. Both can displace mediastinal structures when large. The difference is urgency: most effusions allow time for imaging and fluid analysis, whereas a tension pneumothorax is a bedside diagnosis because cardiovascular collapse can occur while the clinician is waiting for a radiograph.



Pleural fluid forms when normal production/removal balance is overwhelmed

Pleural effusion: presentation and diagnostic approach

Pleural effusions may be asymptomatic when small. When symptomatic, they often cause exertional dyspnea, pleuritic chest discomfort, cough, orthopnea if large, or hypoxemia. Physical examination may show reduced chest expansion, decreased tactile fremitus, dullness to percussion, and diminished breath sounds over the fluid. Above the upper fluid level, compressed lung may transmit bronchial breath sounds or egophony. A large unilateral effusion should trigger a broader differential than bilateral symmetric effusions in a patient with classic volume overload.

Initial imaging usually begins with chest radiography. A posterior-anterior and lateral CXR may show blunting of the costophrenic angle, a meniscus sign, or layering fluid. Roughly 200 to 250 mL of pleural fluid may be needed before a standard upright film reliably shows blunting, whereas ultrasound can detect much smaller volumes and guide safer thoracentesis. Chest CT is useful when malignancy, pulmonary embolism, loculation, pleural thickening, lung abscess, or complicated infection is suspected. Ultrasound is especially useful at the bedside because it can identify septations, mark a safe pocket, and reduce procedure complications.

High-yield principle
New unilateral effusions are usually tapped unless the cause is obvious and the patient is responding as expected. Classic bilateral effusions in decompensated heart failure may be treated first, but thoracentesis becomes appropriate when fever, pleuritic pain, marked asymmetry, lack of response, or another atypical feature is present.

Light criteria and fluid interpretation

The first major branch point is transudate versus exudate. Light criteria are intentionally sensitive for exudates: if any one criterion is positive, the effusion is classified as exudative. This sensitivity matters because missing an exudative effusion may delay diagnosis of pneumonia, malignancy, tuberculosis, pulmonary embolism, or inflammatory disease. The tradeoff is that some heart-failure effusions become falsely exudative after diuresis because water is removed faster than pleural protein. When the clinical story is strongly transudative but Light criteria suggest exudate, clinicians may use serum-pleural albumin gradient or NT-proBNP to support heart failure physiology.

Pleural protein / serum protein > 0.5

Pleural LDH / serum LDH > 0.6

Pleural LDH > 2/3 upper normal serum LDH

If ANY criterion is met, classify as EXUDATIVE

If none are met, the fluid is usually transudative; interpret with clinical context, especially after diuretics.

Pleural fluid test	Typical interpretation	Graph-RAG association
Protein and LDH	High values suggest exudate; Light criteria use protein ratio >0.5, LDH ratio >0.6, or pleural LDH > two-thirds upper serum normal.	exudate -> pleural inflammation / infection / malignancy / PE / TB
pH	A pH <7.20 in suspected pleural infection suggests complicated parapneumonic effusion and need for drainage. Very low pH can also occur with malignancy, rheumatoid pleuritis, esophageal rupture, and TB.	low pH -> impaired glucose metabolism + neutrophilic inflammation + drainage decision
Glucose	Low glucose, often <60 mg/dL and especially <40 mg/dL in infection, suggests high cellular/bacterial metabolism or impaired transport.	low glucose -> empyema / rheumatoid effusion / TB / malignancy / esophageal rupture
Cell differential	Neutrophil-predominant fluid suggests acute inflammation such as pneumonia or PE; lymphocyte predominance suggests TB, malignancy, chronic effusion, or chylothorax.	cells -> timing and cause
Gram stain/culture	Positive results define infected pleural space, but culture may be negative after antibiotics.	microbiology -> antibiotic narrowing + source control
Cytology	High value when malignancy is suspected; sensitivity varies by tumor type and volume submitted.	malignant effusion -> staging + recurrence management
Triglycerides	Elevated triglycerides suggest chylothorax, classically milky fluid from thoracic duct disruption or obstruction.	chyle -> lymphoma / trauma / thoracic surgery
Amylase	Elevated amylase suggests pancreatitis, esophageal rupture, or malignancy.	amylase -> abdominal-esophageal connection

Transudative effusions

Transudative effusions are pressure problems more than pleural problems. The pleural surfaces are not primarily inflamed. In congestive heart failure, elevated venous and pulmonary capillary pressures increase fluid movement into the pleural space. In cirrhosis, ascitic fluid can cross diaphragmatic defects into the chest, often forming a right-sided hepatic hydrothorax. In nephrotic syndrome or severe hypoalbuminemia, low plasma oncotic pressure reduces the ability of the vascular space to hold fluid. Treatment is directed at the systemic cause: diuresis and afterload management for heart failure, sodium restriction and cirrhosis management for hepatic hydrothorax, and treatment of renal or protein-losing disease when applicable.

Therapeutic thoracentesis is not the default for every transudate. If the effusion is small and the patient is minimally symptomatic, treating the underlying disease may be sufficient. If the effusion is massive, causing significant dyspnea, or preventing oxygenation or ventilation, drainage can provide symptomatic relief. Repeated large-volume thoracentesis should be approached carefully because recurrence means the underlying pressure problem persists. Long-term strategies may include optimizing heart failure therapy, addressing cirrhosis, or selected pleural procedures for recurrent symptomatic effusions when definitive systemic control is not possible.

Exudative effusions: infection, malignancy, PE, TB, and autoimmune disease

Exudative effusions occur when the pleural barrier is injured or lymphatic drainage fails. Pneumonia is the classic acute cause. Cancer can obstruct lymphatic drainage, invade pleura, or seed the pleural space. Pulmonary embolism can cause pleuritic inflammation and often produces a small exudative effusion. Tuberculosis classically causes a lymphocyte-predominant exudate and may require pleural biopsy or specific microbiologic testing when suspicion is high. Rheumatoid arthritis can cause a very low glucose effusion; lupus can cause pleuritis with an inflammatory effusion. Pancreatitis can produce an amylase-rich left-sided effusion, and esophageal rupture can cause a toxic pleural process with low pH and high amylase.

The diagnostic approach should be tied to pretest probability. If fever and lobar consolidation are present, the first question is whether this is uncomplicated parapneumonic fluid or a complicated infected space. If weight loss, smoking history, recurrent unilateral effusion, pleural nodularity, or nonresolving opacity is present, cytology and CT evaluation become central. If pleuritic pain and risk factors for venous thromboembolism are present, pulmonary embolism must remain on the list even when the effusion is small. A pleural effusion is not a final diagnosis; it is a fluid expression of another process.

Parapneumonic effusion and empyema

A parapneumonic effusion is pleural fluid associated with pneumonia. Early in infection, sterile inflammatory fluid may collect adjacent to the infected lung; this is an uncomplicated parapneumonic effusion and may resolve with antibiotics. As bacterial burden and neutrophilic inflammation increase, pleural fluid becomes acidic, glucose falls, LDH rises, fibrin deposition creates loculations, and antibiotics penetrate less effectively. At this stage the effusion is complicated and requires drainage. Empyema is frank pus in the pleural space or pleural fluid with organisms, and it requires source control rather than antibiotics alone.

Progression of pleural infection

Pneumonia irritates pleura → Sterile exudative fluid forms → Bacteria/neutrophils enter pleural space → pH falls, glucose falls, LDH rises, loculations form → Drainage plus antibiotics required

Category	Typical fluid/imaging	Treatment logic
Uncomplicated parapneumonic effusion	Free-flowing small-to-moderate fluid; pH usually >=7.20; glucose not severely low; cultures negative.	Antibiotics for pneumonia; observe response. Drain if large, worsening, or respiratory compromise develops.
Complicated parapneumonic effusion	pH <7.20, glucose often <40-60 mg/dL, LDH high, positive Gram stain/culture, or loculated collection.	Antibiotics plus chest tube drainage. Consider intrapleural fibrinolytic/DNase strategies or VATS when drainage is poor.
Empyema	Pus in pleural space or infected pleural fluid. Often fever persists despite antibiotics.	Drainage is essential; antibiotics alone are inadequate. Escalate to VATS/decortication if organized or persistent.

The central management concept is source control. Antibiotics treat pneumonia and bacteremia, but an infected closed space frequently requires drainage. A chest tube is used when the fluid is complicated, infected, purulent, loculated, or causing ongoing sepsis. Surgical consultation is appropriate when tube drainage fails, the lung is trapped by organized pleural peel, or imaging shows complex loculations that cannot be drained adequately. Delayed drainage allows fibrin organization and can convert a manageable infection into chronic fibrothorax with restrictive physiology.

Thoracentesis: indications, risks, and interpretation

Thoracentesis can be diagnostic, therapeutic, or both. Diagnostic thoracentesis identifies the cause of an effusion through chemistry, cell count, microbiology, and cytology. Therapeutic thoracentesis improves dyspnea by reducing mechanical compression of the lung and diaphragm. In most modern practice, ultrasound guidance is preferred because it confirms that fluid is present at the chosen site and helps avoid solid organs, diaphragm, and lung. A very small effusion, especially less than about 10 mm in thickness on decubitus imaging or ultrasound, may be unsafe to tap unless expertise and a compelling reason are present.

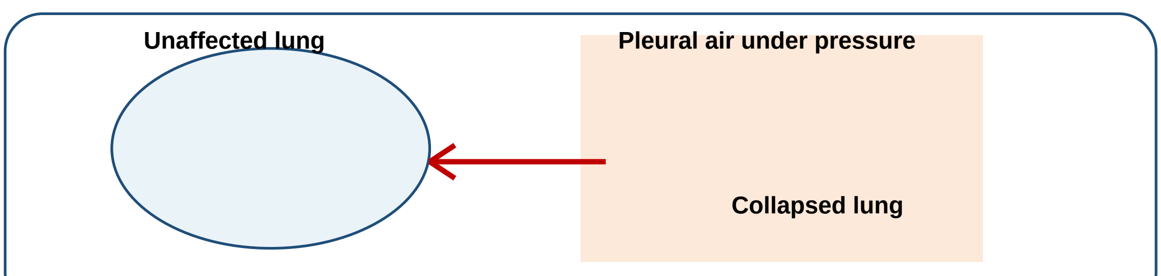
Complications include pneumothorax, bleeding, infection, re-expansion pulmonary edema, pain, cough, and vasovagal symptoms. Pneumothorax risk is reduced by ultrasound guidance, avoiding multiple needle passes, and stopping if air is aspirated. Re-expansion pulmonary edema is uncommon but is classically associated with rapid removal of a large chronic effusion or pneumothorax; it presents with cough, hypoxemia, and infiltrates after drainage. The safest procedural thinking is to define the indication before the needle enters: What result would change management? Is this a diagnostic question, a symptom-relief question, or a source-control question?

When to consider thoracentesis	Why it matters
New unilateral effusion without an obvious cause	Malignancy, infection, PE, TB, and autoimmune disease must be considered.
Effusion with fever, pleuritic pain, or pneumonia	Need to identify complicated parapneumonic effusion or empyema.
Large symptomatic effusion	Drainage can improve dyspnea and oxygenation by restoring lung expansion.
Known CHF/cirrhosis but atypical features	Marked asymmetry, fever, pain, or failure to improve suggests another process.
Suspected malignant effusion	Cytology and staging implications guide oncology and pleural management.
Very small free-flowing effusion without symptoms	Often observe or treat the cause; procedural risk may exceed benefit.

Pneumothorax: classification and physiology

Pneumothorax is air in the pleural space. Because the lung normally stays expanded through negative intrapleural pressure, pleural air uncouples the lung from the chest wall and allows elastic recoil to collapse lung tissue. The clinical impact depends on three variables: size of the air collection, rate of accumulation, and baseline pulmonary reserve. A young patient with a small primary spontaneous pneumothorax may have mild symptoms. A patient with severe COPD and the same size pneumothorax may develop marked hypoxemia because the remaining lung reserve is poor.

Primary spontaneous pneumothorax occurs without known underlying lung disease, classically in tall, thin young adults and often related to rupture of apical subpleural blebs. Secondary spontaneous pneumothorax occurs in the setting of underlying lung disease, including COPD, cystic fibrosis, interstitial lung disease, necrotizing pneumonia, tuberculosis, Pneumocystis infection, lung cancer, or connective tissue disorders. Iatrogenic pneumothorax can follow central venous catheter placement, thoracentesis, lung biopsy, mechanical ventilation, or positive-pressure ventilation. Traumatic pneumothorax follows blunt or penetrating injury and may coexist with hemothorax.



Tension physiology: one-way valve air entry -> ipsilateral collapse + mediastinal shift -> reduced venous return -> shock

Type	Typical patient	Clinical risk	Management tendency
Primary spontaneous	Young patient without known lung disease; often tall/thin; sudden pleuritic pain.	Lower risk if small and stable, but recurrence can occur.	Observation or aspiration/chest tube depending on symptoms and size. Smoking cessation counseling is important.
Secondary spontaneous	COPD, CF, ILD, infection, malignancy, or other lung disease.	Higher risk because pulmonary reserve is limited.	Lower threshold for admission, oxygen, and chest tube drainage.
Iatrogenic	After thoracentesis, line placement, biopsy, ventilation, or procedures.	Depends on size, symptoms, and ventilatory status.	Observe small stable cases; drain if symptomatic, enlarging, or on positive-pressure ventilation.
Traumatic	Blunt/penetrating chest trauma, rib fracture, barotrauma.	May coexist with hemothorax or tension physiology.	Management follows trauma physiology; chest tube often needed.
Tension	Any pneumothorax with rising intrapleural pressure and hemodynamic compromise.	Life-threatening obstructive shock.	Immediate decompression before imaging.

Clinical features and diagnosis of pneumothorax

Symptoms include sudden ipsilateral pleuritic chest pain, dyspnea, cough, anxiety, and sometimes syncope if tension physiology develops. Physical findings may include unilateral decreased breath sounds, hyperresonance to percussion, reduced tactile fremitus, tachycardia, tachypnea, hypoxemia, and subcutaneous emphysema. In a simple pneumothorax, the patient may be stable. In tension pneumothorax, hypotension, distended neck veins, severe respiratory distress, and tracheal deviation away from the affected side can appear, but tracheal deviation is often late and should not be required for action.

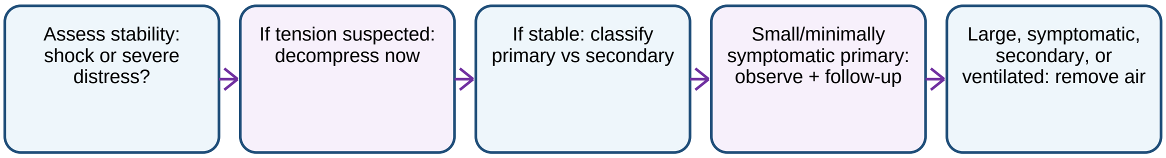
Chest radiography shows a visceral pleural line with absent peripheral lung markings beyond that line. Upright films may show apical pleural air; supine films in ICU or trauma patients may be subtle and can show a deep sulcus sign rather than a classic apical line. Ultrasound can rapidly identify absent lung sliding and a lung point, making it useful at the bedside. CT is the most sensitive test, but CT is not needed before treatment when the patient is unstable. In patients on mechanical ventilation, even a moderate pneumothorax can rapidly worsen under positive pressure.

Management of simple pneumothorax

Management begins with stability. A stable, minimally symptomatic primary spontaneous pneumothorax may be observed with oxygen and repeat imaging, especially if small. Supplemental oxygen can accelerate pleural air resorption by reducing the partial pressure of nitrogen in alveoli and pleural space, although the patient-specific benefit depends on oxygen need and risk of hypercapnia. If the pneumothorax is larger, symptomatic, enlarging, or in a patient with poor reserve, air removal is indicated through needle aspiration, small-bore catheter, or chest tube depending on local practice, severity, recurrence, and underlying disease.

Secondary spontaneous pneumothorax is treated more aggressively because patients with COPD, interstitial lung disease, cystic fibrosis, infection, or malignancy have less reserve. Admission and drainage are more common. Persistent air leak, failure of lung re-expansion, bilateral pneumothoraces, recurrence, or occupational risk may prompt pleurodesis or surgical evaluation. Smoking cessation is a disease-modifying intervention because smoking increases the risk of primary spontaneous pneumothorax and recurrence.

Pneumothorax management ladder



Tension pneumothorax: obstructive shock from pleural air

Tension pneumothorax occurs when air enters the pleural space but cannot escape. Intrapleural pressure rises above atmospheric pressure, collapsing the ipsilateral lung and shifting the mediastinum away. The cardiovascular emergency comes from impaired venous return: the vena cava and right heart cannot fill normally, preload falls, stroke volume falls, and obstructive shock develops. The patient may die from hemodynamic collapse even before hypoxemia is the dominant problem.

The diagnosis is clinical. A patient with severe respiratory distress, hypotension, unilateral absent breath sounds, and risk factors such as trauma, central line placement, or positive-pressure ventilation should be decompressed immediately. Needle decompression or finger thoracostomy is followed by tube thoracostomy because decompression alone does not provide definitive pleural drainage. Waiting for a chest radiograph in an unstable patient is a classic management error. If the patient is stable, imaging can confirm the diagnosis; if unstable and the syndrome fits, the procedure is the diagnostic and therapeutic step.

> **Do-not-miss action**
> Tension pneumothorax is a medical emergency. Treat based on bedside physiology: hypotension + respiratory distress + unilateral decreased breath sounds after trauma/procedure/ventilation is enough to decompress. Imaging is for stable patients, not for delaying care in shock.

Differentials and mimics

Pleural disease often overlaps with parenchymal, vascular, and cardiac disease. Pneumonia can produce consolidation alone, parapneumonic effusion, empyema, lung abscess, or acute respiratory failure. Pulmonary embolism can cause pleuritic pain, dyspnea, small effusion, infarction, hemoptysis, or syncope. Heart failure can cause bilateral effusions and dyspnea but may also coexist with pneumonia or PE. Malignancy can present as recurrent pneumonia from obstruction, pleural effusion from metastasis, or pneumothorax after biopsy. When exam and imaging do not align, avoid anchoring. A patient with decreased breath sounds may have effusion, pneumothorax, mucus plugging, atelectasis, severe hyperinflation, or a large mass.

Finding	Pleural effusion	Pneumothorax	Consolidation/atelectasis
Percussion	Dull over fluid	Hyperresonant over air	Dull if dense consolidation or lobar collapse
Breath sounds	Decreased over effusion; may have bronchial sounds above fluid	Decreased or absent on affected side	Bronchial breath sounds possible in consolidation; decreased in atelectasis
Tactile fremitus	Usually decreased over fluid	Decreased over air	Increased with consolidation; decreased with obstruction/collapse
Imaging clue	Meniscus, blunted angle, layering fluid, ultrasound pocket	Visceral pleural line, absent distal markings, deep sulcus sign supine	Air bronchograms in consolidation; volume loss and shift toward atelectasis
Urgency	Depends on cause and size; empyema requires drainage	Depends on symptoms, size, reserve; tension is immediate	Depends on oxygenation, sepsis, obstruction, and cause

Complications

Pleural effusion complications include progressive hypoxemia, respiratory failure from lung compression, trapped lung, fibrothorax, recurrent symptomatic effusion, and missed malignancy or infection. Empyema complications include sepsis, bronchopleural fistula, loculated infection, pleural peel formation, restrictive lung physiology, and need for surgical decortication. Thoracentesis complications include pneumothorax, bleeding, infection, and re-expansion pulmonary edema. Pneumothorax complications include recurrence, persistent air leak, tension physiology, hypoxemia, and respiratory failure, especially in patients with underlying lung disease or mechanical ventilation.

A useful complication framework is mechanical versus infectious versus diagnostic. Mechanical complications reduce lung expansion or venous return. Infectious complications require source control. Diagnostic complications happen when the effusion is treated as the disease rather than the clue. For example, repeatedly draining a malignant effusion without identifying malignancy misses staging and long-term management; treating a complicated parapneumonic effusion with antibiotics alone misses source control; observing a secondary pneumothorax as if it were primary ignores poor reserve.

High-yield exam clues

Clue	Most useful association
CHF patient with bilateral effusions and no fever	Transudative effusion; treat heart failure first unless atypical.
Pneumonia plus pleural pH <7.20	Complicated parapneumonic effusion; chest tube drainage.
Frank pus from thoracentesis	Empyema; drainage plus antibiotics.
Milky pleural fluid	Chylothorax; consider lymphoma or thoracic duct injury.
Low pleural glucose with rheumatoid arthritis	Rheumatoid pleuritis; can mimic empyema chemically.
High pleural amylase	Pancreatitis, esophageal rupture, or malignancy.
Sudden pleuritic chest pain in tall thin young adult	Primary spontaneous pneumothorax.
COPD patient with acute unilateral absent breath sounds	Secondary pneumothorax until proven otherwise.

Clue	Most useful association
Ventilated trauma patient with hypotension and unilateral absent breath sounds	Tension pneumothorax; decompress immediately.
Trachea deviates away from affected side with shock	Tension pneumothorax or massive effusion; if pneumothorax physiology, do not wait for imaging.

Common pitfalls

Calling every effusion from pneumonia an empyema. Empyema means infected pleural space or pus. An uncomplicated parapneumonic effusion can resolve with antibiotics, but a complicated effusion requires drainage. The difference is not just vocabulary; it changes management.

Using Light criteria without clinical context. Light criteria are sensitive for exudates, but diuresed heart-failure effusions can appear exudative. If the story strongly supports CHF and the patient improves with diuresis, interpret the chemistry carefully rather than reflexively assuming malignancy or infection.

Waiting for imaging in tension pneumothorax. Tension pneumothorax is diagnosed by physiology. Chest radiography can be fatal if it delays decompression in a crashing patient.

Underestimating secondary pneumothorax. A small pneumothorax may be dangerous in severe COPD, interstitial lung disease, cystic fibrosis, or mechanical ventilation. Management depends on reserve, not just size.

Forgetting pulmonary embolism. PE can cause pleuritic pain and a small exudative effusion. If the symptoms are abrupt, the patient has VTE risk factors, or there is unexplained hypoxemia, evaluate for PE even if pleural fluid is present.

Original summary tables for graph-RAG extraction

Concept node	Definition	Key edges
Pleural effusion	Abnormal fluid in pleural space causing lung compression and dyspnea when large.	caused by CHF/cirrhosis/nephrotic syndrome; caused by pneumonia/malignancy/PE/TB; evaluated by CXR/US/thoracentesis; classified by Light criteria
Transudate	Low-inflammatory effusion from systemic pressure or oncotic imbalance.	associated with CHF, cirrhosis, nephrotic syndrome; treated by correcting systemic disease
Exudate	Inflammatory or malignant effusion from pleural injury or impaired lymph drainage.	associated with pneumonia, malignancy, PE, TB, autoimmune disease; requires diagnostic thoracentesis
Light criteria	Three-threshold rule for classifying exudative effusion.	protein ratio >0.5; LDH ratio >0.6; pleural LDH >2/3 upper serum normal; any positive -> exudate
Empyema	Infected pleural space with pus or positive organism evidence.	complication of pneumonia; requires antibiotics plus drainage; may require VATS/decortication
Pneumothorax	Air in pleural space causing partial or complete lung collapse.	primary/secondary/iatrogenic/traumatic; diagnosed by CXR/US; treated by observation/aspiration/chest tube
Tension pneumothorax	Pressurized pleural air causing lung collapse, mediastinal shift, reduced venous return, obstructive shock.	caused by trauma/procedures/ventilation; diagnosed clinically; treated by immediate decompression

Practice questions

1. A 72-year-old man with fever and right lower lobe pneumonia has a moderate pleural effusion. Thoracentesis shows pH 7.12, glucose 32 mg/dL, and LDH 1800 IU/L. What is the next best management step?

Answer explanation: This is a complicated parapneumonic effusion. Antibiotics are necessary but not sufficient. The key next step is chest tube drainage because low pH, low glucose, and high LDH indicate infected/complicated pleural space physiology.

2. A 25-year-old tall, thin man develops sudden right pleuritic chest pain. He is stable, oxygen saturation is 98%, and CXR shows a small apical pneumothorax. What is the likely diagnosis and general management direction?

Answer explanation: Primary spontaneous pneumothorax. If truly small and minimally symptomatic, observation with follow-up imaging may be appropriate; larger or symptomatic cases require air removal.

3. A mechanically ventilated trauma patient suddenly becomes hypotensive and hypoxemic. Breath sounds are absent on the left. What should happen before chest radiography?

Answer explanation: Immediate decompression for suspected tension pneumothorax, followed by definitive tube thoracostomy. The diagnosis is clinical when shock physiology is present.

4. A patient with decompensated cirrhosis has a right-sided pleural effusion. Pleural studies show a transudate and no infection. What mechanism explains this?

Answer explanation: Hepatic hydrothorax: ascitic fluid crosses diaphragmatic defects into the pleural space. It is a pressure/fluid movement problem rather than primary pleural inflammation.

5. A woman with rheumatoid arthritis has a pleural effusion with very low glucose. Why is this potentially confusing?

Answer explanation: Low glucose can suggest empyema, but rheumatoid pleuritis can also produce very low pleural glucose. Clinical context, culture, pH, imaging, and systemic findings help separate inflammatory autoimmune effusion from infected pleural space.

Selected reference anchors

This chapter is original Medvale writing. The uploaded pulmonary reference was used only as a topic coverage checklist for pleural effusion, empyema, pneumothorax, and tension pneumothorax. Current clinical anchors were cross-checked against broad guideline and reference sources: British Thoracic Society Guideline for Pleural Disease 2023; American Association for Thoracic Surgery consensus guidance on empyema; and Merck Manual Professional references on pleural effusion and pneumothorax. These references are included for medical accuracy, not as source text to rewrite.